

HSLU – Hochschule Luzern

T&A – Energy and Environmental Systems Engineering

Laboratory Report
Practical 2
Water Hardness

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1 Introduction

Water hardness is one of the most practically significant parameters in water quality assessment. It is defined as the total concentration of alkaline earth metal ions, primarily calcium (Ca^{2+}) and magnesium (Mg^{2+}), expressed in millimoles per litre (mmol L^{-1}) or as milligrams of calcium carbonate equivalent per litre ($\text{mg L}^{-1} \text{CaCO}_3$). These ions enter water supplies through the dissolution of minerals such as calcite (CaCO_3) and dolomite $\text{CaMg}(\text{CO}_3)_2$ as groundwater percolates through geological formations.

Hardness is classified into two types. Temporary hardness (carbonate hardness) arises from the presence of Ca^{2+} and Mg^{2+} ions associated with bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}) anions. Crucially, this fraction can be removed by boiling, as the following equilibrium is driven to the right upon heating:



Permanent hardness (non-carbonate hardness) is attributable to Ca^{2+} and Mg^{2+} ions paired with sulphate (SO_4^{2-}), chloride (Cl^-), nitrate (NO_3^-), or phosphate (PO_4^{3-}) anions. Because these salts remain in solution even at elevated temperatures, boiling alone is insufficient to remove them.

The principal aim of this experiment is to determine the total hardness of both tap water and commercially bottled water via complexometric titration using the chelating agent ethylenediaminetetraacetic acid (EDTA). EDTA forms stable 1:1 complexes with Ca^{2+} and Mg^{2+} at a buffered pH of approximately 10, enabling accurate quantification of hardness-causing ions. Additionally, the pH behaviour of the sample throughout the titration is recorded to observe the buffering capacity of the water. The results are subsequently compared with the Swiss drinking water quality guidelines and the published values from the Horw municipality water supply. Understanding water hardness is essential for preventing scale deposition in pipework and appliances, optimising soap and detergent consumption, and informing water treatment strategies (World Health Organization, 2011; Stumm & Morgan, 1996).

2 Materials and Methods

2.1 Materials

The following reagents and equipment were used throughout the experiment:

- Water samples: tap water (Horw municipal supply) and commercially bottled water
- 0.01 M EDTA solution (ethylenediaminetetraacetic acid, disodium salt, standard solution)
- Indicator/buffer tablets (ammonium chloride, $\text{NH}_3/\text{NH}_4\text{Cl}$ buffer system, $\times 6$)
- Calibrated pH meter
- 50 mL glass burette with stand and clamp
- 10 mL volumetric pipette
- 100–250 mL beaker ($\times 6$)
- Magnetic stirrer and stirrer bar
- Distilled water for rinsing
- Permanent marker + tape for labelling

2.2 Procedure

All beakers were labelled prior to commencing the experiment. For each titration, 75 mL of the water sample was measured using a volumetric pipette and transferred to a 250 mL beaker. One indicator/buffer tablet was added to the beaker and dissolved completely by stirring. Subsequently, 1 mL of 0.1 M NaOH solution was added to raise the sample pH to approximately 10, as this is the optimal pH range at which EDTA forms stable complexes with Ca^{2+} and Mg^{2+} . A calibrated pH meter was submerged in the sample solution, and the initial pH was recorded.

The burette was rinsed twice with distilled water and once with the EDTA solution before being filled to a suitable starting volume, with the initial reading recorded before each titration, ensuring no air bubbles remained in the tip. The initial burette reading was recorded. EDTA solution was added dropwise (in increments of no more than 0.5 mL) whilst swirling the beaker continuously. The pH was recorded after each 1 mL addition of EDTA. The endpoint was identified by a colour change from brick-red to pure green, indicating complete complexation of all Ca^{2+} and Mg^{2+} ions. The final burette reading was recorded. The procedure was repeated in triplicate for each water sample (three titrations for tap water and three for bottled water).

2.3 Calculation of Water Hardness

Water hardness in $\text{mg L}^{-1} \text{CaCO}_3$ was calculated using the following expression:

$$\text{Hardness } (\text{mg L}^{-1} \text{CaCO}_3) = \frac{(M_{\text{EDTA}} \times V_{\text{EDTA}} \times 1000 \times 100)}{V_{\text{sample}}}$$

Where M_{EDTA} is the molarity of the EDTA solution (mol L^{-1}), V_{EDTA} is the volume of EDTA consumed at the endpoint (mL), V_{sample} is the volume of the water sample (mL), and the constants 1000 and 100 are conversion factors to express the result in mg L^{-1} as CaCO_3 (molar mass of $\text{CaCO}_3 = 100 \text{ g mol}^{-1}$).

To convert the result to French degrees of hardness ($^{\circ}\text{fH}$), the following equivalence was applied:

$$1^{\circ}\text{fH} = 10 \text{ mg L}^{-1} \text{CaCO}_3$$

2.4 Statistical Analysis

Mean, standard deviation (SD), and coefficient of variation ($CV = \sigma / \mu \times 100\%$) were calculated for all quantitative measurements across replicates. Where CVs exceeded 5%, a Grubbs test was applied to the value with the greatest absolute deviation from the mean in each dataset to assess whether it constituted a statistically significant outlier. The test statistic G is calculated as:

$$G = \frac{|x_{suspected} - \bar{x}|}{\sigma}$$

It was then compared against a critical value of 1.155 for $n = 3$ at $\alpha = 0.05$.

3 Results

3.1 Titration Data

Table 1 presents the raw titration data for all six titrations, three replicates per water sample type. EDTA volumes at the endpoint ranged from 8.3 to 10.0 mL for tap water and from 7.3 to 8.7 mL for bottled water, indicating that tap water consistently required greater volumes of EDTA to reach the endpoint, corresponding to a higher concentration of Ca^{2+} and Mg^{2+} ions. Mean endpoint volumes and derived hardness values, with associated standard deviations, are included in the summary rows of Table 1.

Table 1: Titration results for tap and bottled water samples ($n = 3$ per sample type). Mean and SD calculated for the volume of EDTA consumed (V_{EDTA}) and derived water hardness.

Sample Type	V_{SAMPLE} (mL)	Buffer Addition	Initial (mL)	Final (mL)	V_{EDTA} (mL)	Water Hardness ($\text{mg L}^{-1} \text{CaCO}_3$)
Tap Water 1	75	1 tablet	10	20	10	133.3
Tap Water 2	75	1 tablet	40	48.3	8.3	110.7
Tap Water 3	75	1 tablet	20	29.6	9.6	128
Mean $\pm$ SD	75	-	-	-	9.3 ± 0.9	124 ± 11.8
Bottled Water 1	75	1 tablet	30	38.7	8.7	116
Bottled Water 2	75	1 tablet	20	28.5	8.5	113.3
Bottled Water 3	75	1 tablet	30	37.3	7.3	97.3
Mean $\pm$ SD	75	-	-	-	8.2 ± 0.8	108.9 ± 10.1

3.2 pH Measurements During Titration

Tables 2 and 3 record the pH of each sample at 1 mL increments of EDTA addition, up to the individual endpoint volume for each replicate. Across both sample types, the pH remained stable throughout the titration: tap water averages declined from 9.1 to 8.8 (a total change of 0.3 pH units) and bottled water averages declined from 8.9 to 8.7 (0.2 pH units). This near-constant profile reflects the strong buffering capacity of the $\text{NH}_3/\text{NH}_4\text{Cl}$ system, reinforced by the 1 mL NaOH addition, which maintained conditions at approximately pH 9–10 throughout. Notably, Sample 2 of tap water displayed consistently higher pH readings (9.4–9.5) compared to Samples 1 and 3, suggesting a possible difference in initial sample conditions for that replicate.

Table 2: pH as a function of volume of 0.01 M EDTA added. Values in parentheses indicate the actual endpoint volume where it does not correspond to the row V_{EDTA} value. Mean, SD, and CV calculated for tap water samples.

V_{EDTA} (mL)	Sample 1	Sample 2	Sample 3	Mean $\pm$ SD	CV (%)
0	9	9.5	8.8	9.1 ± 0.4	4.4
1	9	9.5	8.8	9.1 ± 0.4	4.4

V_{EDTA} (mL)	Sample 1	Sample 2	Sample 3	Mean $\pm$ SD	CV (%)
2	9	9.5	8.8	9.1 ± 0.4	4.4
3	8.9	9.5	8.7	9 ± 0.4	4.4
4	8.9	9.5	8.7	9 ± 0.4	4.4
5	8.9	9.5	8.7	9 ± 0.4	4.4
6	8.9	9.5	8.7	9 ± 0.4	4.4
7	8.9	9.5	8.7	9 ± 0.4	4.4
8	8.8	9.5	8.7	9 ± 0.4	4.4
9	8.8	(8.3 mL) 9.4	8.7	9 ± 0.4	4.4
10	8.8	-	(9.6 mL) 8.7	8.8 ± 0.1	1.1

Table 3: pH as a function of volume of 0.01 M EDTA added. Values in parentheses indicate the actual endpoint volume where it does not correspond to the row V_{EDTA} value. Mean, SD, and CV calculated for bottled water samples.

V_{EDTA} (mL)	Sample 1	Sample 2	Sample 3	Mean $\pm$ SD	CV (%)
0	8.9	8.9	9	8.9 ± 0.1	1.1
1	8.9	8.9	9	8.9 ± 0.1	1.1
2	8.8	8.9	9	8.9 ± 0.1	1.1
3	8.8	8.9	8.9	8.9 ± 0.1	1.1
4	8.8	8.9	8.9	8.9 ± 0.1	1.1
5	8.8	8.9	8.8	8.8 ± 0.1	1.1
6	8.8	8.8	8.8	8.8 ± 0	0
7	8.8	8.8	8.8	8.8 ± 0	0
8	8.7	8.8	(7.3 mL) 8.8	8.8 ± 0.1	1.1
9	(8.7 mL) 8.7	(8.7 mL) 8.8	-	8.8 ± 0.1	1.1

3.3 pH vs. Volume Graph

Figure 1 plots the mean pH profile for both water types against the volume of EDTA added, with individual replicate series shown as dashed lines and endpoint volumes marked by triangular markers. The near-flat trajectories for both sample types confirm the strongly buffered nature of the system; no inflection point was observable, and endpoint detection relied entirely on the indicator colour change from brick-red to green. The tap water average sits consistently 0.2–0.3 pH units above the bottled water average throughout, consistent with the higher initial ionic strength of the tap water sample.

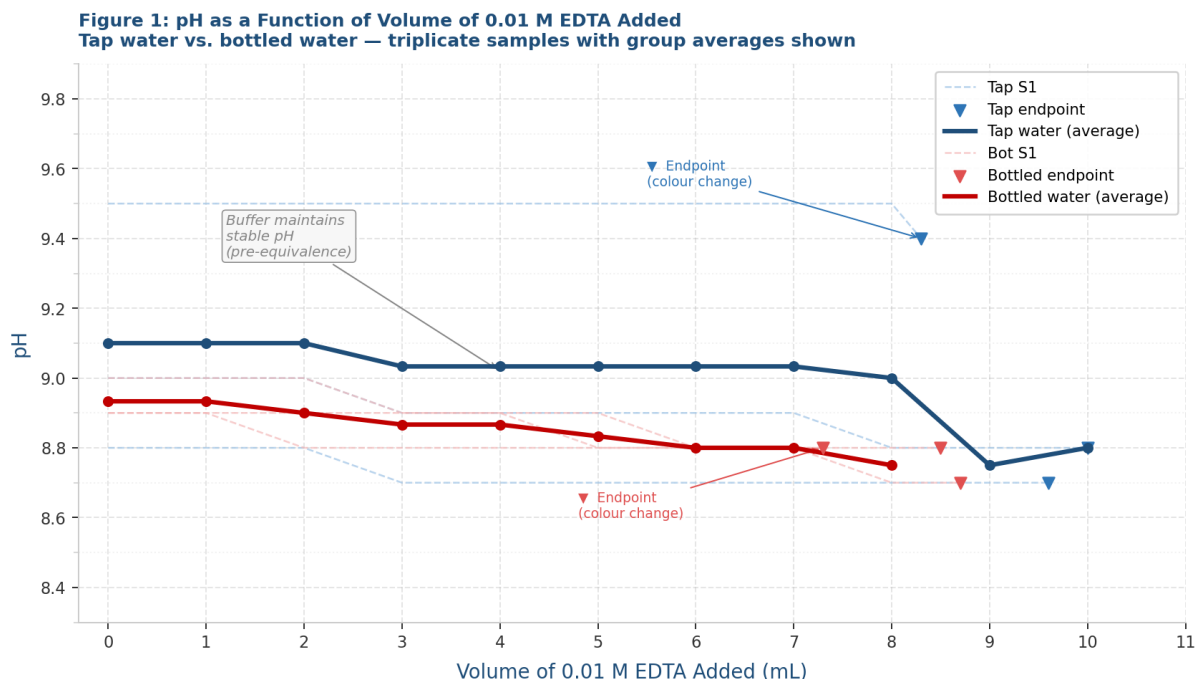


Figure 1: pH as a function of volume of 0.01 M EDTA added for tap water and bottled water samples ($n = 3$ per type). Triangular markers ($\blacktriangledown$) indicate the endpoint volume for each individual replicate. Buffer: $\text{NH}_3/\text{NH}_4\text{Cl}$ tablet (containing indicator) + 1 mL NaOH .

3.4 Sample Calculation

The following worked example, using the equation defined in 2.3, illustrates the full calculation pathway from EDTA volume to hardness in both $\text{mg L}^{-1} \text{CaCO}_3$ and French degrees ($^\circ\text{fH}$), using the mean V_{EDTA} across the three tap water replicates:

$$\text{Hardness } (\text{mg L}^{-1} \text{CaCO}_3) = \frac{(0.01 \times 9.3 \times 1000 \times 100)}{75} = 124 \text{ mg L}^{-1} \text{CaCO}_3$$

Converting to French degrees:

$$124 \text{ mg L}^{-1} \div 10 = 12.4 \text{ }^\circ\text{fH} \Rightarrow \text{classified as Soft (7 – 15 }^\circ\text{fH)}$$

Applying the same formula to bottled water (mean $V_{\text{EDTA}} = 8.2 \text{ mL}$):

$$\begin{aligned} \text{Hardness } (\text{mg L}^{-1} \text{CaCO}_3) &= \frac{(0.01 \times 8.2 \times 1000 \times 100)}{75} = 109 \text{ mg L}^{-1} \text{CaCO}_3 \\ &\Rightarrow 10.9 \text{ }^\circ\text{fH} \Rightarrow \text{Soft (7 – 15 }^\circ\text{fH)} \end{aligned}$$

3.5 Statistical Analysis

Table 4 summarises the descriptive statistics for the three replicates per sample type. The mean and median values for tap water (124 ± 11.8 and $128 \text{ mg L}^{-1} \text{CaCO}_3$ respectively) diverge slightly, suggestive of a mildly left-skewed distribution, with the mean falling below the median, consistent with the comparatively low value of Sample 2 (110.7 mg L^{-1}). The coefficient of variation for tap water is 9.5% and for bottled water 9.3%, both exceeding the 5% threshold generally considered acceptable for analytical titrations. As both CVs exceed this threshold, a Grubbs test was applied to determine whether the most extreme value in each dataset constitutes a statistical outlier.

For tap water, the value with the greatest absolute deviation from the mean is Sample 2 (110.7 mg L⁻¹): $G = 1.13$, below the critical value of 1.155, and therefore not a statistically significant outlier. For bottled water, the most extreme value is Sample 3 (97.3 mg L⁻¹): $G = 1.15$, also marginally below the critical value. Thus, no replicates were excluded. The proximity of both G statistics to the critical threshold, combined with the near-identical CVs across both sample types, indicates that the variability is likely systematic in origin rather than sample-specific, and is discussed further in 4.6.

Table 4: Descriptive statistics for water hardness ($n = 3$ per sample type) derived from Table 1, and mean endpoint pH averaged across the three replicates for each sample type.

Sample Type	Mean V_{EDTA} (mL)	Mean $\pm$ SD (mg L ⁻¹ CaCO ₃)	Median (mg L ⁻¹ CaCO ₃)	CV (%)	Mean pH
Tap Water	9.3	124 $\pm$ 11.8	128	9.5	9
Bottled Water	8.2	108.9 $\pm$ 10.1	113.3	9.3	8.8

3.6 Hardness Classification

Table 5 classifies the mean hardness values for both sample types against the Swiss hardness scale (Kanton Solothurn, 2008). Both tap water (12.4 °fH) and bottled water (10.9 °fH) fall within the Soft category (7–15 °fH), which is below the typical Swiss municipal range of 14–18 °fH. The implications of this classification are discussed in 4.2.

Table 5: Water hardness classification in French degrees (°fH) with experiment results (Kanton Solothurn, 2008).

Water Hardness (°fH)	mmol L ⁻¹	Description	Result
0 to 7	0 to 0.7	Very soft	
7 to 15	0.7 to 1.5	Soft	Tap + Bottled samples
15 to 25	1.5 to 2.5	Moderately hard	
25 to 32	2.5 to 3.2	Fairly hard	
32 to 42	3.2 to 4.2	Hard	

4 Discussion

4.1 Comparison with Municipal Water Quality Data (Horw)

The mean tap water hardness of $124 \pm 11.8 \text{ mg L}^{-1} \text{ CaCO}_3$ (12.4 °fH) falls below the Swiss national range of 14–18 °fH (140–180 $\text{mg L}^{-1} \text{ CaCO}_3$; Stumm & Morgan, 1996), yet aligns closely with the published Horw municipality value of 11.8 °fH (Trinkwasser.ch, 2025). This agreement suggests the titration results are valid and that any remaining discrepancy is attributable to minor analytical error rather than a systematic methodological flaw. The relatively soft character of Horw municipal water is consistent with its primary source: Lake Lucerne, which is fed predominantly by alpine snowmelt and has limited contact time with carbonate-bearing geology (primarily limestone and dolomite), resulting in naturally low Ca^{2+} and Mg^{2+} concentrations. This illustrates that the national average is not representative of all Swiss municipalities: the Biregg area of Horw, supplied by Energie Wasser Luzern (EWL) from a separate mixed water source, exhibits notably harder water due to greater limestone contact (Gemeinde Horw, n.d.), demonstrating how local geology and source type drive hardness variability at the sub-municipal level.

Although the declared mineral content of the bottled water label was unavailable for direct comparison, the mean hardness of $108.9 \pm 10.1 \text{ mg L}^{-1} \text{ CaCO}_3$ (10.9 °fH) marginally exceeds the WHO minimum palatability recommendation of 100 mg L^{-1} (WHO, 2011). The lower hardness relative to tap water suggests the bottled water originates from a source with comparably limited mineral contact, possibly also a surface or glacially-fed spring, though without label data this remains speculative.

4.2 Classification of Results

Both tap water (12.4 °fH) and bottled water (10.9 °fH) fall within the Soft classification (7–15 °fH), sitting in the lower half of that band. From a practical standpoint, water at this hardness level lathers readily with soap, poses negligible risk of limescale accumulation in appliances, and requires no softening treatment for domestic or most industrial uses. However, the soft classification is not without drawback. Unlike moderately hard water, which deposits a thin protective CaCO_3 layer on the internal surfaces of pipes, soft water lacks this passivating film and is consequently more corrosive towards metallic pipework, potentially contributing to elevated copper or lead concentrations at the tap in older plumbing systems (Stumm & Morgan, 1996). Both results also comfortably exceed the Very Soft threshold (below 7 °fH), beneath which corrosion risk increases sharply and the water may have an unpleasant, flat taste due to the absence of dissolved minerals.

4.3 Formation of Hard Water and Mitigation Strategies

Hard water is formed when naturally soft precipitation water, which is slightly acidic due to dissolved atmospheric CO_2 forming carbonic acid (H_2CO_3), percolates through calcareous rock formations such as limestone and dolomite. The following dissolution reaction illustrates the key process:



The longer groundwater remains in contact with these minerals, the greater the concentration of dissolved Ca^{2+} and Mg^{2+} ions. In regions where groundwater has extended contact with calcareous geography, this process produces moderately to fairly hard drinking water, as reflected in the Swiss national average of 14–18 °fH. However, as noted in 4.1, national characterisation does not apply

uniformly: regions fed by alpine lakes and snowmelt tend to produce softer water than areas with more limestone contact.

Several strategies can mitigate or remove water hardness: (1) Ion exchange softeners replace Ca^{2+} and Mg^{2+} with Na^+ ions using zeolite or synthetic resin beds; (2) Reverse osmosis membranes remove dissolved ions by forcing water under pressure through a semi-permeable membrane; (3) Chemical precipitation using lime ($\text{Ca}(\text{OH})_2$) and soda ash (Na_2CO_3) converts dissolved hardness ions into insoluble carbonates; (4) Boiling removes temporary (carbonate) hardness by driving off CO_2 and precipitating CaCO_3 . Permanent hardness, however, requires chemical or membrane treatment as it cannot be removed by heating alone.

4.4 pH Behaviour During EDTA Titration

As demonstrated by Figure 1, the pH profiles recorded during this experiment did not exhibit the sigmoidal inflection typically associated with EDTA titrations in textbook descriptions. Instead, both sample types displayed near-constant pH throughout: a total decline of 0.3 pH units for tap water and 0.2 pH units for bottled water, consistent with a strongly buffered system. The $\text{NH}_3/\text{NH}_4\text{Cl}$ buffer tablet, combined with the 1 mL NaOH addition, maintained pH between 8.7 and 9.5 throughout, effectively suppressing the pH variation that would otherwise accompany progressive metal ion complexation.

This buffering is essential for the chemistry of the titration itself: the conditional stability constant of EDTA–metal ion complexes is highly pH-dependent, and at pH 9–10, EDTA binds effectively with both Ca^{2+} and Mg^{2+} , whereas at lower pH values EDTA becomes increasingly protonated and unavailable for complexation (Skoog et al., 2014). Maintaining this pH window therefore ensures complete and selective complexation of hardness ions, which is the chemical basis for the accuracy of the method.

Before the equivalence point, incoming EDTA immediately binds with free Ca^{2+} and Mg^{2+} ions, so no measurable change in free EDTA concentration occurs and the pH remains stable. At the equivalence point, all hardness ions have been chelated by EDTA; the metallochromic indicator, a compound that forms a coloured complex with metal ions and displays a distinctly different colour when free in solution, is displaced by EDTA, which has a greater binding affinity for Ca^{2+} and Mg^{2+} , releasing the indicator into its free form and producing the observed colour change from brick-red to green. After the equivalence point, excess EDTA accumulates in solution, but the buffer continues to suppress any pH shift, explaining the continued flatness of the curve beyond the endpoint.

As a consequence of the strong buffering, no equivalence point could be potentiometrically identified from the pH data alone, and endpoint detection relied entirely on the indicator colour change. This underscores a fundamental limitation of visual endpoint detection in buffered chelatometric titrations: had the buffer been less effective, a pH-based endpoint would have offered greater objectivity and reproducibility, reducing the inter-replicate variability observed in this experiment.

4.5 Significance of Water Hardness Measurement

The determination of water hardness is of considerable importance across multiple domains. From a domestic perspective, hard water reacts with soaps and detergents to form insoluble calcium and magnesium stearate precipitates (soap scum), reducing cleaning efficiency and increasing product consumption. In household appliances such as washing machines, dishwashers, and water heaters, carbonate scale (limescale) gradually deposits on heating elements, reducing thermal efficiency and ultimately shortening appliance lifetimes.

From an industrial standpoint, scale deposition in boilers, heat exchangers, and cooling towers represents a significant economic burden due to reduced heat transfer and increased maintenance costs. In the food and beverage industry, water hardness directly affects the taste and texture of products. At 12.4 and 10.9 °fH respectively, both tap and bottled water samples sit above the Very Soft threshold (below 7 °fH) where corrosion risk increases, but below the Moderately Hard range where scaling becomes problematic, making this an ideal range for drinking water from an infrastructure perspective.

Environmentally, elevated hardness in wastewater discharge can influence freshwater ecosystem chemistry. The hardness-to-alkalinity ratio governs the solubility of heavy metals and nutrient availability in aquatic environments. Regarding human health, moderate hardness (approximately 75–100 mg L⁻¹ CaCO₃) has been associated with reduced cardiovascular disease risk in epidemiological studies, as Ca²⁺ and Mg²⁺ are essential micronutrients (WHO, 2011). Conversely, very hard water may contribute to urolithiasis (kidney stone formation) in susceptible individuals due to elevated urinary calcium excretion. The WHO guideline does not specify a maximum hardness limit for health reasons but recommends a minimum of 100 mg L⁻¹ for palatability.

4.6 Statistical Analysis and Discussion of Results

Both CVs exceeded the 5% threshold generally considered acceptable in analytical chemistry titrations, 9.5% for tap water and 9.3% for bottled water, indicating that the triplicates did not achieve satisfactory reproducibility. As detailed in 3.5, neither the tap water ($G = 1.13$) nor the bottled water ($G = 1.15$) extreme values constituted statistical outliers at $\alpha = 0.05$.

The consistency of both G statistics approaching but not reaching the critical threshold, combined with the near-identical CVs for both sample types, strongly suggests that the variability is systematic in origin rather than sample-specific. The most probable source is inconsistent endpoint detection: the indicator colour transition from brick-red to green is gradual rather than sharp, and under variable laboratory lighting conditions, different group members may have identified the endpoint at slightly different points in the transition. This is supported by the pH data, which showed that no potentiometric confirmation of the endpoint was available, leaving visual judgement as the sole determinant. Secondary sources of error may include parallax in burette readings, particularly relevant given that readings were taken at multiple volumes across each titration.

Despite the CVs exceeding the 5% threshold, the final results of 124 ± 11.8 mg L⁻¹ CaCO₃ (12.4 °fH) for tap water and 108.9 ± 10.1 mg L⁻¹ CaCO₃ (10.9 °fH) for bottled water are internally consistent with one another and align well with the published Horw municipal value of 11.8 °fH (Trinkwasser.ch, 2025), suggesting the method produced directionally accurate results despite the reproducibility limitations.

5 Conclusion

This experiment quantified the total water hardness of tap water and commercially bottled water via EDTA complexometric titration, yielding mean hardness values of $124 \pm 11.8 \text{ mg L}^{-1} \text{ CaCO}_3$ (12.4 °fH) and $108.9 \pm 10.1 \text{ mg L}^{-1} \text{ CaCO}_3$ (10.9 °fH) respectively. Both results fall within the Soft classification (7–15 °fH) and are consistent with the published Horw municipal hardness of 11.8 °fH (Trinkwasser.ch, 2025), supporting the validity of the titration method despite CVs exceeding the 5% reproducibility threshold.

A key concept reinforced by this experiment is the carbonic acid–carbonate equilibrium, which governs the dissolution of limestone minerals into groundwater and drives the natural formation of water hardness. The practical consequence of this chemistry is directly observable at the municipal level: the softness of Horw's water reflects the dominance of alpine snowmelt as a source, contrasting with harder supplies derived from greater limestone contact, such as the EWL-supplied Biregg area.

The primary sources of imprecision were the subjective visual detection of the endpoint under variable lighting conditions, the possibility of overshooting the EDTA addition beyond the true endpoint before the colour change was recognised, parallax in burette readings, and the absence of a potentiometric inflection point due to strong buffering, which prevented objective confirmation of the endpoint. These factors account for the elevated CVs observed and suggest that future work employing automated spectrophotometric or potentiometric endpoint detection would substantially improve reproducibility.

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Declaration of AI

AI (Claude, Anthropic) was used throughout the preparation of this report. Specific uses include: structuring and drafting sections of the introduction, Discussion and Conclusion; identifying inconsistencies and errors in referencing; and proofreading for grammar and register. All scientific content and experimental data are the work of the authors. Scientific interpretations were reviewed, revised and approved by the authors, and the final submitted text represents the author's own understanding of the material.